Chapter 11

Campbell Biology, AP Edition, 12e

A cheetah chases an impala.

Chapter 11********

Cell Communication

Figure 11.1

This impala is fleeing for its life, racing to escape the predatory cheetah nipping at its heels. The impala is breathing rapidly, its heart pounding and its legs pumping furiously. These physiological functions are all part of the impala's "fightor- flight" response, driven by hormones released from its adrenal glands at times of stress—in this case, upon sensing the cheetah.

How does cell signaling fuel the desperate flight of an impala?

The impala senses a cheetah. Its brain signals the adrenal glands to release epinephrine into the blood. Signal reception. An epinephrine molecule binds to a receptor on a muscle cell. Signal transduction. Relay molecules transmit the signal, ultimately activating an enzyme. Cellular response. The enzyme breaks down glycogen, releasing glucose that fuels the leg muscles.

Key Concepts

- **11.1External signals are converted to responses within the cell****
- **11.2Signal reception: A signaling molecule binds to a receptor, causing it to change shape****
- **11.3Signal transduction: Cascades of molecular interactions transmit signals from receptors to relay molecules in the cell****

- **11.4Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities****
- **11.5Apoptosis requires integration of multiple cell-signaling pathways****

Advanced placement edition

Mechanisms for cell-to-cell communication have been preserved across domains, for both unicellular and multicellular organisms **(BIG IDEA 1)**. Cells can communicate via direct contact or through signaling molecules that act locally or over long distances **(BIG IDEA 3)**.

Study Tip

Make a table: As you read about examples of cell signaling in this chapter, make a table and classify the events of each example into three stages: signal reception, signal transduction, and cellular response.

A table lists an example of cell signaling, signal reception, signal transduction, and cellular response.

The table is as follows:

Example of cell signaling	Signal reception	Signal transduction	Cellular response
Epinephrine.	Epinephrine binds cell-surface receptor.	Relay molecules each activate the next molecule.	An enzyme is activated that breaks down glycogen into glucose for energy to fight or flee.

Get Ready for This Chapter

In this chapter, you will learn the three steps involved in cell-cell signaling: reception, transduction, and response. Before beginning this chapter, you should be able to do the following:

- Explain how proteins bind to each other (see Concept 2.3 and Figure 2.16).
- Describe how hydrolysis of ATP drives transport (see Concept 7.4 and Figure 7.16).
- Describe allosteric binding (see Concept 8.5 and Figure 8.20).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which property is necessary for protein A to bind to protein B?

Protein A must have regions of shape and charge that are complementary to those on Protein B.

That's correct.

Correct! If you had trouble with this question, review the following material: Molecular shape determines how biological molecules recognize and respond to one another with specificity. Biological molecules often bind temporarily to each other by forming weak interactions, but only if their shapes are complementary, and if their functional groups and surface charges are compatible. The role of molecular shape illustrates how biological organization leads to a match between structure and function, one of biology's unifying themes.

```
        Any two molecules can bind to each other; there is no
special property required.
```

That's incorrect. Try again.

Incorrect. Review the following material: Molecular shape determines how biological molecules recognize and respond to one another with specificity. Biological molecules often bind temporarily to each other by forming weak interactions, but only if their shapes are complementary, and if their functional groups and surface charges are compatible. The role of molecular shape illustrates how biological organization leads to a match between structure and function, one of biology's unifying themes.

Protein A and Protein B must be identical so they can form a pair.

That's incorrect. Try again.

Incorrect. Review the following material: Molecular shape determines how biological molecules recognize and respond to one another with specificity. Biological molecules often bind temporarily to each other by forming weak interactions, but only if their shapes are complementary, and if their functional groups and surface charges are compatible. The role of molecular shape illustrates how biological organization leads to a match between structure and function, one of biology's unifying themes.

Protein A must have regions of shape and charge that are the same as those on Protein B.

That's incorrect. Try again.

Incorrect. Review the following material: Molecular shape determines how biological molecules recognize and respond to one another with specificity. Biological molecules often bind temporarily to each other by forming weak interactions, but only if their shapes are complementary, and if their functional groups and surface charges are compatible. The role of molecular shape illustrates how biological organization leads to a match between structure and function, one of biology's unifying themes.

2. How does hydrolysis of ATP drive transport?

ATP hydrolysis moves protons across the membrane and the proton-motive force is used to transport substances.

That's incorrect. Try again.

Incorrect. Review the following material: One way ATP hydrolysis can power active transport is when its terminal phosphate group is transferred directly to the transport protein. This can induce the protein to change its shape in a manner that translocates a solute bound to the protein across the membrane. The transport proteins that move solutes against their concentration gradients are all carrier proteins rather than channel proteins.

The transport protein synthesizes ATP as it transports ions down their concentration gradient across the membrane.

That's incorrect. Try again.

Incorrect. Review the following material: One way ATP hydrolysis can power active transport is when its terminal phosphate group is transferred directly to the transport protein. This can induce the protein to change its shape in a manner that translocates a solute bound to the protein across the membrane. The transport proteins that move solutes against their concentration gradients are all carrier proteins rather than channel proteins.

A phosphate group from ATP is added to a transport protein, changing its shape and facilitating transport.

That's correct.

Correct! If you had trouble with this question, review the following material: One way ATP hydrolysis can power active transport is when its terminal phosphate group is transferred directly to the transport protein. This can induce the protein to change its shape in a manner that translocates a solute bound to the protein across the membrane. The transport proteins that move solutes against their concentration gradients are all carrier proteins rather than channel proteins.

Transport cannot be driven by ATP hydrolysis—substances are transported through channels in channel proteins.

That's incorrect. Try again.

Incorrect. Review the following material: One way ATP hydrolysis can power active transport is when its terminal phosphate group is transferred directly to the transport protein. This can induce the protein to change its shape in a manner that translocates a solute bound to the protein across the membrane. The

transport proteins that move solutes against their concentration gradients are all carrier proteins rather than channel proteins.

3. You have isolated a previously unstudied protein, identified its complete structure in detail, and determined that it is an enzyme that catalyzes the breakdown of a large substrate. You notice it has two binding sites: one large and one small. Speculate about how this enzyme might function.

```
        The enzyme catalyzes two different reactions, each with
its own specific substrate.
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That's incorrect. Try again.

Incorrect. Review the following material: Allosteric regulation is the term used to describe any case in which a protein's function at one site is affected by the binding of a small regulatory molecule to a separate site. In the simplest kind of allosteric regulation, an activating or inhibiting regulatory molecule binds to a regulatory site (sometimes called an allosteric site). The binding of an activator to a regulatory site stabilizes the shape that has functional active sites, whereas the binding of an inhibitor stabilizes the inactive form of the enzyme.

```
        The enzyme works through competitive inhibition—
different substrates compete for binding to each of the two
sites.
```

That's incorrect. Try again.

Incorrect. Review the following material: Allosteric regulation is the term used to describe any case in which a protein's function at one site is affected by the binding of a small regulatory molecule to a separate site. In the simplest kind of allosteric regulation, an activating or inhibiting regulatory molecule binds to a regulatory site (sometimes called an allosteric site). The binding of an activator to a regulatory site stabilizes the shape that has functional active sites, whereas the binding of an inhibitor stabilizes the inactive form of the enzyme.

```
        It is probably a cell membrane transport protein—like an
ion channel.
```

That's incorrect. Try again.

Incorrect. Review the following material: Allosteric regulation is the term used to describe any case in which a protein's function at one site is affected by the binding of a small regulatory molecule to a separate site. In the simplest kind of allosteric regulation, an activating or inhibiting regulatory molecule binds to a regulatory site (sometimes called an allosteric site). The binding of an activator to a regulatory site stabilizes the shape that has functional active sites, whereas the binding of an inhibitor stabilizes the inactive form of the enzyme.

```
        The small site is probably a binding site for a small
molecule whose binding affects the enzyme's shape and function—
its breakdown of the large substrate.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Allosteric regulation is the term used to describe any case in which a protein's function at one site is affected by the binding of a small regulatory molecule to a separate site. In the simplest kind of allosteric regulation, an activating or inhibiting regulatory molecule binds to a regulatory site (sometimes called an allosteric site). The binding of an activator to a regulatory site stabilizes the shape that has functional active sites, whereas the binding of an inhibitor stabilizes the inactive form of the enzyme.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 11.1Describe how external signals are converted to responses within cells.
- 11.2Identify different types of signal receptors and explain how they function.
- 11.3 Explain the roles of phosphorylation, dephosphorylation, and second messengers in signal transduction.
- 11.4Describe how cells respond to signaling, including fine-tuning of the response.
- 11.5Use apoptosis as an example to illustrate how multiple signaling pathways can be integrated in a cell.

Concept 11.1: External signals are converted to responses within the cell

Scientists think that signaling mechanisms first evolved hundreds of millions of years ago in ancient prokaryotes and single-celled eukaryotes and then were adapted for new uses in their multicellular descendants. So let's begin by considering signaling in some examples of single-celled organisms: bacteria and yeasts.

Evolution of Cell Signaling

EVOLUTION Research during the 1970s suggested that bacterial cells—somewhat surprisingly, since they are single-celled organisms—were capable of signaling to each other. Since then, we have come to understand that cell signaling is critical among prokaryotes. Bacterial cells secrete molecules that can be detected by other bacterial cells **(Figure 11.2)**. Sensing the concentration of such signaling molecules allows bacteria to monitor their own local cell density, a phenomenon called *quorum sensing*.

Figure 11.2

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Communication among bacteria.
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Communication among bacteria. Step 1. Individual rod-shaped cells on a 0.5 millimeters scale signal. step 2. Aggregation in progress. Step 3. Spore-forming structure (fruiting body) is formed on a 2.5 millimeters scale.

Soil-dwelling bacteria called myxobacteria ("slime bacteria") use chemical signals to share information about nutrient availability. When food is scarce, starving cells secrete a signaling molecule that stimulates neighboring cells to aggregate. The cells form a structure called a fruiting body that produces spores, thick-walled cells that can survive until the environment improves. The myxobacteria shown here are the species Myxococcus xanthus (steps 1–3, SEMs; lower photo, LM).

Quorum sensing allows bacterial populations to coordinate the behavior of all cells in a population in activities that require a given density of cells acting at the same time. One example is formation of a *biofilm*, an aggregation of bacterial cells attached to a surface by molecules secreted by the cells, but only after the cells have reached a certain density. The biofilm protects the cells in it, and they often derive nutrition from the surface they are on. Biofilms are believed to be involved in up to 80% of all human bacterial infections.

You have probably encountered biofilms many times, perhaps without realizing it. The slimy coating on a fallen log or on leaves lying on a forest path, and even the film on your teeth each morning, are examples of bacterial biofilms. In fact, tooth-brushing and flossing disrupt biofilms that would otherwise cause cavities and gum disease.

HHMI BioInteractive Video

Interview with Bonnie Bassler

image

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image

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For more resources from HHMI BioInteractive, go to www.biointeractive.org.

BBC Video

Brushing Your Teeth Can Save Your Life

image

Play

image

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Interview

Interview with Bonnie Bassler: Exploring how bacteria communicate with each other

A photo of Bonnie Bassier.

Another example of bacterial behavior coordinated by quorum sensing is the secretion of toxins by infectious bacteria, which has serious medical implications. Sometimes treatment by antibiotics doesn't work with such infections because of antibiotic resistance that has evolved in a particular strain of bacteria. A promising alternative treatment would be to disrupt toxin production by interfering with the signaling pathways used in quorum sensing. In the **Problem-Solving Exercise**, you can participate in the process of scientific thinking involved in this novel approach.

Problem-Solving Exercise

Can a skin wound turn deadly?

AP

A chart asks, can a skin wound turn deadly?

Staphylococcus aureus (S. aureus) is a common bacterial species found on the surface of healthy skin that can turn into a serious pathogen if introduced into the tissue through a cut or abrasion. Once S. aureus cells enter the body and reach a certain density, they secrete a toxin that kills body cells and contributes significantly to inflammation and damage. Because about one in 100 people carry a strain of S. aureus that is resistant to common antibiotics, a minor infection can turn permanently harmful or even deadly.

Staphylococcus aureus (S. aureus) is a common bacterial species found on the surface of healthy skin that can turn into a serious pathogen if introduced into tissue through a cut or abrasion. Once *S. aureus* cells enter the body and reach a certain density, they secrete a toxin that kills body cells and contributes significantly to inflammation and damage. Because about one in 100 people carry a strain of *S. aureus* that is resistant to common antibiotics, a minor infection can turn permanently harmful or even deadly.

Cells sense their own population density by *quorum sensing*, and at a certain density they start to secrete toxin. In this exercise, you will analyze whether blocking quorum sensing can stop *S. aureus* from producing toxin.

Your Approach In *S. aureus,* quorum sensing involves two separate signal transduction pathways. Two candidate synthetic peptides (short proteins), called peptides 1 and 2, have been proposed to interfere with *S. aureus* quorum-sensing pathways. Your job is to test each potential inhibitor of quorum sensing to see if it blocks either or both of the pathways that lead to toxin production.

For your experiment, you grow four cultures of *S. aureus* to a standardized high density and measure the concentration of toxin in the culture. The control culture contains no peptide. The other cultures have one or both candidate inhibitory peptides mixed into the growth medium before starting the cultures.

Your Data

Bar graph. The x-axis shows concentrations for four cultures: control, peptide 1, peptide 2, and peptides 1 plus 2. The y-axis has concentration of toxin in culture (micromoles per milliliter) and ranges from 0 to 2.5 in increments of 0.5 units.

Data from N. Balaban et al., Treatment of *Staphylococcus aureus* biofilm infection by the quorum-sensing inhibitor RIP, *Antimicrobial Agents and Chemotherapy* 51(6):2226–2229 (2007).

Control: 2.25 micromoles per milliliter. Peptide 1: 1.0 micromole per milliliter. Peptide 2: 0.75 micromoles per milliliter. Peptides 1 plus 2 show the lowest concentration 0.2 micromoles per milliliter.

Your Analysis

- Rank the cultures according to toxin production, from most to least.
- Which, if any, of the cultures with peptide(s) resulted in a toxin concentration similar to the control culture? What is your evidence for this?
- Was there an additive effect on toxin production when peptides 1 and 2 were both present in the growth medium? What is your evidence for this?
- Based on these data, would you hypothesize that peptides 1 and 2 act on the same quorum-sensing pathway leading to toxin production or on two different pathways? What is your reasoning?
- Do these data suggest a possible treatment for antibiotic-resistant *S. aureus* infections? What else would you want to know to investigate this further?

Instructors: A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**. Or a more extensive investigation called “Solve It: Is It Possible to Treat Bacterial Infections Without Traditional Antibiotics?” can be assigned.

Now let’s look at an example of cell signaling in yeasts (single-celled fungi). Cells of the yeast *Saccharomyces cerevisiae*—which are used to make bread, wine, and beer—identify their sexual mates by chemical signaling when they reproduce sexually. There are two sexes, or mating types, called a and α **(Figure 11.3)**. Each type secretes a specific factor that binds only to receptors on the other type of cell. When exposed to each other’s mating factors, a pair of cells of opposite type change shape, grow toward each other, and fuse (mate). The new a/α cell contains all the genes of both original cells, providing advantages to the cell’s descendants, which arise by subsequent cell divisions.

Figure 11.3

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Communication between mating yeast cells.
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Communication between yeast cells.

Step 1. Exchange of mating factors. Each mating cell type secretes a mating factor that binds to receptors on the other mating type.

Step 2. Mating. Binding of the factors to receptors induces changes in the cells that lead to their fusion. Alpha and a cells are now fused, but retain distinct nuclei.

Step 3. New a, alpha cell. The nucleus of the fused cell includes all the genes from the a and alpha cells. The fused cells from step 2 have become one cell with one nucleus containing all the genes from both the a and alpha cells.

Saccharomyces cerevisiae cells use chemical signaling to identify cells of the opposite mating type and initiate the mating process. The two mating types and their corresponding chemical signaling molecules, or mating factors, are called a and α.

The unique match between mating factor and receptor is key to ensuring mating only between cells of the same species of yeast. How does the binding of a mating factor by the yeast cell-surface receptor initiate a signal that brings about the cellular response of mating? This occurs in a series of three major steps—signal reception, signal transduction, and cellular response—called a *signal transduction pathway*. Many such pathways exist in the cells of both unicellular and multicellular organisms. In fact, the molecular details of signal transduction in yeasts and mammals are strikingly similar, even though it's been over a billion years since they shared a common ancestor. This similarity suggests that early versions of cell-signaling mechanisms evolved well before the first multicellular organisms appeared on Earth.

Local and Long-Distance Signaling

Like bacteria or yeast cells, cells in a multicellular organism communicate by way of a signaling molecules targeted for cells that may or may not be immediately adjacent. As we saw in Concepts 6.7 and 7.1, eukaryotic cells may communicate by direct contact, which is one type of local signaling. Many animal and plant cells have cell junctions that directly connect the cytoplasms of adjacent cells **(Figure 11.4a)**. In these cases, signaling substances dissolved in the cytosol can pass between neighboring cells. Moreover, some animal cells may communicate by direct contact between cell-surface molecules, as shown in **Figure 11.4b**. This type of local signaling is especially important in embryonic development, the immune response, and in maintaining adult stem cell populations.

Figure 11.4

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Communication requiring contact between cells.
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Communication requiring contact between cells.

Part a. Cell junctions. Both animals and plants have cell junctions that allow molecules, including signaling molecules, to pass readily between adjacent cells without crossing plasma membranes. Gap junctions between animal cells and plasmodesmata between plant cells allow signaling substances to pass between the cytoplasm of adjacent cells.

Part b. Cell-cell recognition. Two cells in an animal may communicate by interaction between molecules protruding from their surfaces. Two animal cells with different receptors on their surfaces bind at the receptors, allowing cell signaling to occur.

In many other cases of local signaling, signaling molecules are secreted by the signaling cell. Some molecules travel only short distances; such local regulators influence cells that are nearby. This type of local signaling in animals is called *paracrine signaling***(Figure 11.5a)**. One class of local regulators in animals, *growth factors*, consists of compounds that stimulate nearby target cells to grow and divide. Numerous cells can simultaneously receive and respond to the growth factors produced by a single cell in their vicinity.

Figure 11.5

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    Local and long-distance cell signaling by secreted molecules
in animals.
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Local and long-distance cell signaling by secreted molecules in animals.

In both local and long-distance signaling, only specific target cells that can recognize a given signaling molecule will respond to it.

Part a. Local signaling: Paracrine signaling. A signaling cell acts on nearby target cells by secreting molecules of a local regulator (a growth factor, for example). Secreting cell with secretory vessels secretes a local regulatory molecule that can act on nearby target cells.

Part b. Local signaling: Synaptic signaling. A nerve cell releases neurotransmitter molecules into a synapse, stimulating the target cell, such as a muscle or another nerve cell. An electrical signal triggers the release of neurotransmitters, which then diffuse across the synapse (space between nerve cell and target cell) to the target cell.

Part C. Long-distance signaling: Endocrine (hormonal) signaling. Specialized endocrine cells secrete hormones into body fluids, often blood. Hormones reach most body cells, but are bound by and affect only some cells. A human endocrine cell releases hormones that travel in a blood vessel to a specific target cell that can bind the hormone.

A highly specialized type of local signaling called *synaptic signaling* occurs in the animal nervous system **Figure 11.5b**; see Concept 48.4. An electrical signal along a nerve cell triggers the secretion of neurotransmitter molecules. These molecules act as chemical signals, diffusing across the synapse—the narrow space between the nerve cell and its target cell—triggering a response in the target cell. Drugs to treat depression, anxiety, and post-traumatic stress disorder (PTSD) affect this signaling process.

Both animals and plants use molecules called **hormones** for long-distance signaling. In hormonal signaling in animals, also known as *endocrine signaling*, specialized cells release hormones, which travel through the circulatory system to other parts of the body, where they reach target cells that can recognize and respond to them **(Figure 11.5c)**. Many plant hormones reach distant targets by traveling through cells (see Concept 39.2). Like local regulators, plant hormones vary widely in size and type. For instance, the plant hormone ethylene, which promotes fruit ripening and helps regulate growth, is a hydrocarbon of only six atoms (C_2H_4), small enough to diffuse through the air and pass through cell walls. (Small signaling molecules probably evolved early on among single-celled organisms.) In contrast, the mammalian hormone insulin, which regulates sugar levels in the blood, is a protein with thousands of atoms.

What happens when a potential target cell is exposed to a secreted signaling molecule? The ability of a cell to respond is determined by whether it has a specific receptor molecule that can bind to the signaling molecule. The information conveyed by this binding, the signal, must then be changed into another form—transduced—inside the cell before the cell can respond. The remainder of the chapter discusses this process, primarily as it occurs in animal cells. (Two signaling pathway proteins are shown in their cellular context in Figure 6.32a.)

The Three Stages of Cell Signaling: *A Preview*

Our current understanding of how signaling molecules act through signal transduction pathways had its origins in the pioneering work of Earl W. Sutherland, whose research led to a Nobel Prize in 1971. Sutherland and his colleagues at Vanderbilt University were investigating how the animal hormone epinephrine (also called adrenaline) triggers the “fight-or-flight” response in animals like the impala in Figure 11.1. One effect of epinephrine is to mobilize fuel reserves, which can be used by the animal to either defend itself (fight) or try to escape (flight). Epinephrine stimulates the breakdown of the storage polysaccharide glycogen within liver cells and skeletal muscle cells. The breakdown of glycogen releases the sugar glucose 1-phosphate, which the cell converts to glucose 6-phosphate. The liver or muscle cell can then use this compound, an early intermediate in glycolysis, for energy production (see Figure 9.8). Alternatively, the compound can be stripped of phosphate and released from the liver cell into the blood as glucose, which can fuel cells throughout the body.

But how, exactly, does epinephrine mobilize glucose for use? Sutherland's research team discovered that epinephrine outside the cell stimulates glycogen breakdown by somehow activating an enzyme, glycogen phosphorylase, inside the cell. However, when epinephrine was added to a solution containing the enzyme and its substrate, glycogen, no breakdown occurred. Glycogen phosphorylase could be activated by epinephrine only when the hormone was added to intact cells. This result told Sutherland two things. First, epinephrine does not interact directly with the enzyme responsible for glycogen breakdown; an intermediate step or series of steps must be occurring in the cell. Second, an intact, membrane-bound cell must be present for transmission of the signal to take place.

Sutherland's work suggested that the process going on at the receiving end of a cellular communication can be dissected into three stages: signal reception, signal transduction, and cellular response **(Figure 11.6)**:

- ******STEP 1:****Signal reception.** Reception is the target cell's detection of a signaling molecule coming from outside the cell. A chemical signal is "detected" when the signaling molecule binds to a receptor protein located at the cell's surface (or inside the cell, to be discussed later).

- ******STEP 2:****Signal transduction.** The binding of the signaling molecule changes the receptor protein in some way, initiating the process of transduction. The transduction stage converts the signal to a form that can bring about a specific cellular response. In Sutherland's system, the binding of epinephrine to a receptor protein in a liver cell's plasma membrane leads to activation of glycogen phosphorylase in the cytosol. Transduction sometimes occurs in a single step but more often requires a sequence of changes in a series of different molecules—a **signal transduction pathway**. The molecules in the pathway are often called relay molecules; three are shown as an example.

- ******STEP 3:****Cellular response.** The transduced signal finally triggers a specific cellular response. The response may be almost any imaginable cellular activity—such as catalysis by an enzyme (for example, glycogen phosphorylase), rearrangement of the cytoskeleton, or activation of specific genes in the nucleus. The cell-signaling process helps ensure that crucial activities like these occur in the right cells, at the right time, and in proper coordination with the activities of other cells of the organism. We'll now explore the mechanisms of cell signaling in more detail, including a discussion of regulation and termination of the process.

Figure 11.6

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Overview of cell signaling.
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An overview of cell signaling.

From the perspective of the cell receiving the message, cell signaling can be divided into three stages: signal reception, signal transduction, and cellular response. When reception occurs at the plasma membrane, as shown here, the transduction stage is usually a pathway of several steps (three are shown as an example), with each specific relay molecule in the pathway bringing about a change in the next molecule. The final molecule in the pathway triggers the cell's response.

Binding of a signaling molecule from the extracellular fluid to a receptor in the plasma membrane triggers reception, transduction, and response within the cell cytoplasm.

1. Signal reception. The signaling molecule in the extracellular fluid binds to a receptor in the plasma membrane.
2. Signal transduction. In this diagram, three relay molecules show a signal transduction pathway where each relay molecule brings about a change in the next molecule in the pathway.
3. Cellular response. The final molecule in the pathway in step 2 activates a cellular response.

VISUAL SKILLS*Where would the epinephrine in Sutherland's experiment fit into this diagram of cell signaling?*

*For suggested answer, see*Appendix A.**

Animation

Overview of Cell Signaling

image

Play

image

Play

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Concept 11.2: Signal reception: A signaling molecule binds to a receptor, causing it to change shape

A wireless router may broadcast its network signal indiscriminately, but only computers with the correct password can connect to it: Reception of the signal depends on the receiver. Similarly, the signals emitted by an a mating type yeast cell are “heard” only by its prospective mates, α cells. In the case of the epinephrine circulating throughout the bloodstream of the impala in Figure 11.1, the hormone encounters many types of cells, but only certain target cells, those with the corresponding receptor protein, detect and respond to the epinephrine molecule. The signaling molecule is complementary in shape to a specific site on the receptor and attaches there, like a hand in a glove. The signaling molecule acts as a **ligand**, the term for a molecule that specifically binds to another (often larger) molecule. Ligand binding generally causes a receptor protein to undergo a change in shape. For many receptors, this shape change directly activates the receptor, enabling it to interact with other molecules in or on the cell. For other receptors, the immediate effect of ligand binding is to cause the aggregation of two or more receptor proteins, which leads to further molecular events inside the cell. Most signal receptors are plasma membrane proteins, but others are located inside the cell. We’ll look at both of these types next.

Animation

Reception

image

Play

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Receptors in the Plasma Membrane

Cell-surface transmembrane receptors play crucial roles in the biological systems of animals. The largest family of human cell-surface receptors is that of the **G protein-coupled receptors(GPCRs)**. There are more than 800 GPCRs; an example is shown in **Figure 11.7**. Another example is the co-receptor hijacked by

HIV to enter immune cells (see Figure 7.8); this GPCR is the target of the drug maraviroc, which has shown some success at treating AIDS.

Figure 11.7

The structure of a G protein-coupled receptor (GPCR).

A computer model of the human beta 2 adrenergic receptor within the plasma membrane. Within it is cholesterol and a molecule that mimics a ligand.

This is a model of the human β2-adrenergic receptor, which binds epinephrine (adrenaline). The receptor was crystallized (discussed later in this section) in the presence of both a molecule that mimics epinephrine (green in the model) and cholesterol in the membrane (orange). Two receptors (blue) are shown as ribbon models in a side view. Caffeine can also bind to this receptor; see question 10 at the end of the chapter.

Most water-soluble signaling molecules bind to specific sites on transmembrane receptor proteins that transmit information from the extracellular environment to the inside of the cell. We can see how cell-surface transmembrane receptors work by looking at three major types: G protein-coupled receptors (GPCRs), receptor tyrosine kinases, and ion channel receptors. These receptors are discussed and illustrated in **Figure 11.8**; study this figure before going on.

Figure 11.8

Exploring Cell-Surface Transmembrane Receptors

A diagram of G protein-coupled receptors, receptor tyrosine kinases, and ion channel receptors.

G protein-coupled receptors: A G protein-coupled receptor (G P C R) is a cell-surface transmembrane receptor that works with the help of a G protein, a protein that binds the energy-rich molecule G T P. Many different signaling molecules—including yeast mating factors, neurotransmitters, and epinephrine (adrenaline) and many other hormones—use G P C R s. G protein-coupled receptors vary in the binding sites for their ligands and also for different types of G proteins inside the cell. Nevertheless, G P C R proteins are all remarkably similar in structure. In fact, they make up a large family of eukaryotic receptor proteins with a secondary structure in which the single polypeptide, represented here in a ribbon model, has seven transmembrane alpha helices (outlined with cylinders and depicted in a row for clarity). Specific loops between the helices (here, the loops on the right) form binding sites for signaling molecules (outside the cell) and G proteins (on the cytoplasmic side). G P C R-based signaling systems are extremely widespread and diverse in their functions, including roles in embryonic development and sensory reception. In humans, for example, vision, smell, and taste depend on G P C R s (see Concept 50.4). Similarities in structure in G proteins and G P C R s in diverse organisms suggest that G proteins and their associated receptors evolved

very early among eukaryotes. Malfunctions of the associated G proteins themselves are involved in many human diseases, including bacterial infections. The bacteria that cause cholera, pertussis (whooping cough), and botulism, among others, make their victims ill by producing toxins that interfere with G protein function. Up to 60% of all medicines used today exert their effects by influencing G protein pathways. Figure describes G protein-coupled receptors in four steps:

- Loosely attached but able to move along the cytoplasmic side of the membrane, a G protein functions as a molecular switch that is either on or off, depending on whether G D P or G T P is attached—hence the term G protein. (G T P, or guanosine triphosphate, is similar to A T P.) When G D P is bound to the G protein, as shown above, the G protein is inactive. The receptor and G protein work together with another protein, usually an enzyme.
- When the appropriate signaling molecule binds to the extracellular side of the receptor, the receptor is activated and changes shape. Its cytoplasmic side then binds an inactive G protein, causing a G T P to displace the G D P. This activates the G protein.
- The activated G protein dissociates from the receptor, diffuses along the membrane, and then binds to an enzyme, altering the enzyme's shape and activity. Once activated, the enzyme can trigger the next step leading to a cellular response. Binding of signaling molecules is reversible: Like other ligands, they bind and dissociate many times. The ligand concentration outside the cell determines how often a ligand is bound and initiates signaling.
- The changes in the enzyme and G protein are only temporary because the G protein also functions as a G T Pase enzyme—in other words, it then hydrolyzes its bound G T P to G D P. Now inactive again, the G protein leaves the enzyme, which returns to its original state. The G protein is now available for reuse. The G T Pase function of the G protein allows the pathway to shut down rapidly when the signaling molecule is no longer present.

MAKE CONNECTIONS *Is the flow of ions through a ligand-gated channel an example of active or passive transport? (Review**Concepts 7.3**and*7.4.)*

*For suggested answer, see*Appendix A.**

Animation

Acetylcholine Receptor

image

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Play

Given the many important functions of cell-surface receptors, it is not surprising that their malfunctions are associated with many human diseases, including cancer, heart disease, and asthma. To better understand and treat these conditions, a major focus of both university research teams and the pharmaceutical industry has been to analyze the structure of these receptors.

Although cell-surface receptors (half of which are GPCRs) represent 30% of all human proteins, determining their structures by X-ray crystallography (see Figure 5.21) has proved challenging. For one thing, cell-surface receptors tend to be flexible and inherently unstable, thus difficult to crystallize. It took years of persistent efforts for researchers to determine the first few of these structures, such as the GPCR shown in Figure 11.7. In that case, the β-adrenergic receptor was stable enough to be crystallized only while it was among membrane molecules and in the presence of a molecule mimicking its ligand. Newer techniques that do not require crystallization, like cryo-EM (see Figure 6.3), have shown promise in determining the structures of cell-surface receptors.

Abnormal functioning of receptor tyrosine kinases (RTKs) is associated with many types of cancers. For example, some breast cancer patients have tumor cells with excessive levels of a receptor tyrosine kinase called HER2 (see Concept 12.3 and Figure 18.27). Using molecular biological techniques, researchers have developed a protein called Herceptin that binds to HER2 on cells and inhibits cell division, thus thwarting further tumor development. In some clinical studies, treatment with Herceptin improved patient survival rates by more than one-third. One goal of ongoing research into these cell-surface receptors and other cell-signaling proteins is development of additional successful treatments.

Interview

Interview with Diana Bautista: How cell signaling leads to itch and pain

Diana Bautista.

Intracellular Receptors

Intracellular receptor proteins are found in either the cytoplasm or nucleus of target cells. To reach such a receptor, a signaling molecule passes through the target cell's plasma membrane. A number of important signaling molecules can do this because they are either hydrophobic enough or small enough to cross the hydrophobic interior of the membrane (see Concept 7.1). Hydrophobic signaling molecules include steroid hormones and thyroid hormones of animals. Another chemical signaling molecule that possesses an intracellular receptor is nitric oxide (NO), a gas; this very small molecule readily passes between the

membrane phospholipids. Once a hormone or other signaling molecule has entered a cell, its binding to an intracellular receptor changes the receptor into a hormone-receptor complex that is able to cause a response —in many cases, the turning on or off of particular genes.

The behavior of aldosterone is a representative example of how steroid hormones work. This hormone is secreted by cells of the adrenal gland (a gland that lies above the kidney), then travels through the blood and enters cells all over the body. However, a response occurs only in kidney cells because they alone contain receptors for aldosterone. In these cells, the hormone binds to and activates the receptor protein. With aldosterone attached, the active form of the receptor protein then enters the nucleus and turns on specific genes that control water and sodium flow in kidney cells, ultimately affecting blood volume **(Figure 11.9)**.

Figure 11.9

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Steroid hormone interacting with an intracellular receptor.
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The process of steroid hormone interaction in five steps.

Step 1. The steroid hormone aldosterone passes through the plasma membrane.

Step 2. Aldosterone binds to a receptor protein in the cytoplasm, activating it.

Step 3. The hormone-receptor complex enters the nucleus and binds to specific genes.

Step 4. The bound protein acts as a transcription factor, stimulating the transcription of the gene into m R N A.

Step 5. The m R N A is translated into a specific protein.

MAKE CONNECTIONS *Why is a cell-surface receptor protein not required for this steroid hormone to enter the cell? (See*Concept 7.2.)**

*For suggested answer, see*Appendix A.**

Animation

Steroid Hormone Pathway

image

Play

image

Play

**

How does the activated hormone-receptor complex turn on genes? Recall that the genes in a cell's DNA function by being transcribed and processed into messenger RNA (mRNA), which leaves the nucleus and is translated into a specific protein by ribosomes in the cytoplasm (see Figure 5.22). Specialized proteins called *transcription factors* control which genes are turned on—that is, which genes are transcribed into mRNA—in a particular cell at a particular time. When the aldosterone receptor is activated, it acts as a transcription factor that turns on specific genes. (You'll learn more about transcription factors in Chapters 17 and 18.)

By acting as a transcription factor, the aldosterone receptor itself carries out two parts of the signaling pathway, as receptor and transducer. Most other intracellular receptors function in the same way, although many of them, such as the thyroid hormone receptor, are already in the nucleus before the signaling molecule reaches them. Interestingly, many of these intracellular receptor proteins are structurally similar, suggesting an evolutionary kinship.

Concept 11.3: Signal transduction: Cascades of molecular interactions transmit signals from receptors to relay molecules in the cell

When receptors for signaling molecules are plasma membrane proteins, like most of those we have discussed, the transduction stage of cell signaling is usually a multistep pathway involving many molecules. Steps often include activation of proteins by addition or removal of phosphate groups or release of other small molecules or ions that act as signaling molecules. One benefit of using multiple steps is that a signal caused by a small number of signaling molecules can be greatly amplified. If each molecule transmits the signal to numerous molecules at the next step in the series, the result is a geometric increase in the number of activated molecules by the end (see Figure 11.16). A second benefit of using multistep pathways is that they provide more opportunities for coordination and control than do simpler systems. This allows regulation of the response, as we'll see later in the chapter.

Signal Transduction Pathways

The binding of a specific signaling molecule to a receptor in the plasma membrane triggers the first step in the signal transduction pathway—the chain of molecular interactions that leads to a particular response

within the cell. Like falling dominoes, the signal-activated receptor activates another molecule, which activates yet another molecule, and so on, until the protein that produces the final cellular response is activated. The molecules that relay a signal from receptor to response, which we call relay molecules in this book, are often proteins. Protein-protein interactions are a major theme of cell signaling—indeed, a unifying theme of all cellular activities.

Animation

Signal Transduction Pathways

image

Play

image

Play

Keep in mind that the original signaling molecule is not physically passed along a signaling pathway; in most cases, it never even enters the cell. When we say that the signal is relayed along a pathway, we mean that certain information is passed on. At each step, the signal is transduced into a different form, commonly a shape change in the next protein. Very often, the shape change is brought about by phosphorylation.

Protein Phosphorylation and Dephosphorylation

Previous chapters introduced the concept of activating a protein by adding one or more phosphate groups to it (see Figure 8.11a). In Figure 11.8, you have already seen how phosphorylation is involved in the activation of receptor tyrosine kinases. In fact, phosphorylation of proteins and its reverse, dephosphorylation, are commonly used in cells to regulate protein activity. An enzyme that transfers phosphate groups from ATP to a protein is generally known as a **protein kinase**. Recall that a receptor tyrosine kinase (RTK) is a specific kind of protein kinase that phosphorylates tyrosines on the other RTK in a dimer. Most cytoplasmic protein kinases, however, act on proteins different from themselves. Another distinction is that most cytoplasmic protein kinases phosphorylate either of two other amino acids, serine or threonine, rather than tyrosine. Serine/threonine kinases are widely involved in signaling pathways in animals, plants, and fungi.

Many signal transduction pathways use relay molecules that are protein kinases, and they often act on other protein kinases in the pathway. **Figure 11.10** depicts a hypothetical pathway containing two different protein kinases that create a **phosphorylation cascade**. The sequence of steps shown in the figure is similar to many known pathways, including those triggered in yeast by mating factors and in animal cells by many growth factors. The signal is transmitted by a cascade of protein phosphorylations, each causing a

shape change in the phosphorylated protein. The shape change results from the interaction of the newly added phosphate groups with charged or polar amino acids on the protein being phosphorylated (see Figure 5.14). The shape change in turn alters the function of the protein, most often activating it. (In some cases, though, phosphorylation instead *decreases* the activity of the protein.)

Figure 11.10

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A phosphorylation cascade.
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A phosphorylation cascade in four steps.

In a phosphorylation cascade, a series of different proteins in a pathway are phosphorylated in turn, each protein adding a phosphate group to the next one in line. Here, phosphorylation activates each protein, and dephosphorylation returns it to its inactive form. The active and inactive forms of each protein are represented by different shapes to remind you that activation is usually associated with a change in molecular shape.

A signaling molecule binds to a receptor in the plasma membrane, triggering a series of events inside the cell. Steps 2 through 4 below are part of the phosphorylation cascade.

- An activated relay molecule activates protein kinase 1.
- Active protein kinase 1 transfers a phosphate group from A T P (leaving A D P) to an inactive molecule of protein kinase 2, activating this second kinase.
- Active protein kinase 2 phosphorylates a protein that brings about the cell‘s response to the signal.
- Protein phosphatases (P P) catalyze the removal of the phosphate groups from the proteins, making the proteins inactive again and available for reuse.

WHAT IF *What would happen if a mutation in protein kinase2made it incapable of being phosphorylated?*

*For suggested answer, see*Appendix A.**

A significant percentage of our own genes—about 2%—is thought to code for protein kinases. A single cell may have hundreds of different kinds, each specific for a different substrate protein. Together, protein kinases probably regulate the activity of a large proportion of the thousands of proteins in a cell. Among these are most of the proteins that, in turn, regulate cell division. Abnormal activity of such a kinase can cause abnormal cell division and contribute to the development of cancer.

Equally important in the phosphorylation cascade are the **protein phosphatases**, enzymes that can rapidly remove phosphate groups from proteins, a process called *dephosphorylation*. By dephosphorylating and thus inactivating protein kinases, phosphatases provide the mechanism for turning off the signal transduction pathway when the initial signal is no longer present. Phosphatases also make the protein kinases available for reuse, enabling the cell to respond again to an extracellular signal. The phosphorylation-dephosphorylation system acts as a molecular switch in the cell, turning activities on or off, or up or down, as required. At any given moment, the activity of a protein regulated by phosphorylation depends on the balance in the cell between active kinase molecules and active phosphatase molecules.

Small Molecules and Ions as Second Messengers

Not all components of signal transduction pathways are proteins. Many signaling pathways also involve small, nonprotein, water-soluble molecules or ions called **second messengers**. (The pathway's "first messenger" is considered to be the extracellular signaling molecule—the ligand—that binds to the membrane receptor.) Because they are small and water-soluble, they can readily spread through regions of the cell by diffusion. For example, as we'll see shortly, a second messenger called cyclic AMP carries the signal initiated by epinephrine from the plasma membrane of a liver or muscle cell into the cell's interior, where the signal eventually brings about glycogen breakdown. Second messengers participate in pathways that are initiated by both G protein-coupled receptors and receptor tyrosine kinases. The two most common second messengers are cyclic AMP and calcium ions, Ca^{2+}. A large variety of relay proteins respond to changes in the cytosolic concentration of one or the other of these second messengers.

Cyclic AMP

Having discovered that epinephrine somehow causes glycogen breakdown within cells, Earl Sutherland next looked for a second messenger that transmits the signal from the plasma membrane to the metabolic machinery in the cytoplasm. He found that binding of epinephrine to the G protein-coupled receptor (GPCR) in the plasma membrane results in a rise in the cytosolic concentration of **cyclic AMP(cAMP**; cyclic adenosine monophosphate), a small molecule produced from ATP. As shown in **Figure 11.11**, an enzyme embedded in the plasma membrane, **adenylyl cyclase** (also known as adenylate cyclase), converts ATP to cAMP in response to an extracellular signal—in this case, provided by epinephrine. When epinephrine outside the cell binds to a GPCR, it activates a G protein that in turn activates adenylyl cyclase. Adenylyl cyclase can then catalyze the synthesis of many molecules of cAMP. In this way, the normal cellular concentration of cAMP can be boosted 20-fold in a matter of seconds. The cAMP broadcasts the signal to the cytoplasm. It does not persist for long in the absence of the hormone because a different enzyme, called

phosphodiesterase, converts cAMP to AMP. Another surge of epinephrine is needed to boost the cytosolic concentration of cAMP again.

Figure 11.11

Cyclic AMP.

Cyclic AMP.

A T P is split into pyrophosphate and c AMP by the enzyme Adenylyl cyclase.

The structure of A T P is a tail of three phosphate groups (a central P with a double bond to one O and single bonds to three other O's; each group is bonded together between an O and the central P of the next group) attached to carbon 5 of the sugar ribose, which is attached to adenine (a nitrogen-containing double ring structure). Pyrophosphate is two phosphate groups joined together. c AMP is a phosphate group attached to carbons 5 and 3 of ribose, which is attached to adenine. c AMP can go to AMP by hydrolysis with the enzyme phosphodiesterase. AMP is one phosphate group attached to ribose which is attached to adenine.

The second messenger cyclic AMP (cAMP) is made from ATP by adenylyl cyclase, an enzyme embedded in the plasma membrane. The phosphate group in cAMP is attached to both the 5′ and the 3′ carbons; cAMP's cyclic arrangement is the basis for its name. cAMP is inactivated by phosphodiesterase, an enzyme that converts it to AMP.

WHAT IF? *What would happen if a molecule that inactivated phosphodiesterase were introduced into the cell?*

*For suggested answer, see*Appendix A.**

Subsequent research has revealed that epinephrine is only one of many hormones and many other signaling molecules that lead to activation of adenylyl cyclase by G proteins and formation of cAMP **(Figure 11.12)**. The immediate effect of an elevation in cAMP level is usually the activation of a serine/threonine kinase called *protein kinase A*. The activated protein kinase A then phosphorylates various other proteins, depending on the cell type. (The complete pathway for epinephrine's stimulation of glycogen breakdown is shown later, in Figure 11.16.)

Figure 11.12

```
cAMP as a second messenger in a G protein signaling pathway.
```

c AMP as a second messenger in a G protein signaling pathway.

The first messenger is a signaling molecule such as epinephrine.

- First messenger binds to G P C R, activating it.
- Activated G P C R binds to G protein, which is then bound by G T P, activating the G protein.
- Activated G protein/ G T P binds to adenylyl cyclase. G T P is hydrolyzed, activating adenylyl cyclase.
- Activated adenylyl cyclase converts A T P to c AMP.
- c AMP, a second messenger, activates another protein, leading to cellular responses.

Further regulation of cell metabolism is provided by other G protein systems that *inhibit* adenylyl cyclase. In these systems, a different signaling molecule activates a different receptor, which in turn activates an *inhibitory* G protein that blocks activation of adenylyl cyclase. Cell activities can be fine-tuned by the balance between these systems.

Now that we know about the role of cAMP in G protein signaling pathways, we can explain in molecular detail how certain microorganisms cause disease. Consider cholera, a disease that is frequently epidemic in places where the water supply is contaminated with human feces. People acquire the cholera bacterium, *Vibrio cholerae*, by drinking contaminated water. The bacteria form a biofilm on the lining of the small intestine and produce a toxin. The cholera toxin is an enzyme that chemically modifies a G protein involved in regulating salt and water secretion. Because the modified G protein is unable to hydrolyze GTP to GDP, it remains stuck in its active form, continuously stimulating adenylyl cyclase to make cAMP (see the question with Figure 11.12). The resulting high concentration of cAMP causes the intestinal cells to secrete large amounts of salts into the intestines, with water following by osmosis. An infected person quickly develops profuse diarrhea and if left untreated can soon die from the loss of water and salts.

Our understanding of signaling pathways involving cyclic AMP or related messengers has allowed us to develop treatments for certain conditions in humans such as erectile dysfunction. In one pathway, the gas nitric oxide NO is released by a cell and enters a neighboring muscle cell, where it causes production of a molecule similar to cAMP called *cyclic GMP* (*cGMP*). The cGMP then acts as a second messenger that causes relaxation of muscles, such as those in the walls of arteries. A compound that prolongs the signal (by inhibiting the hydrolysis of cGMP to GMP) was originally prescribed for chest pains because it relaxed blood vessels and increased blood flow to the heart muscle. Under the trade name Viagra, this compound is now widely used as a treatment for erectile dysfunction in human males. Because Viagra leads to dilation of blood vessels, it also allows increased blood flow to the penis, optimizing physiological conditions for penile erections.

Calcium Ions and Inositol Trisphosphate (IP3)

Many of the signaling molecules that function in animals—including neurotransmitters, growth factors, and some hormones—induce responses in their target cells through signal transduction pathways that increase the cytosolic concentration of calcium ions (Ca^{2+}). Calcium is even more widely used than cAMP as a second messenger. Increasing the local cytosolic concentration of Ca^{2+} causes many responses in animal cells, including muscle cell contraction, exocytosis of molecules (secretion), and cell division. In plant cells, a wide range of hormonal and environmental stimuli can cause brief increases in cytosolic Ca^{2+} concentration, triggering various signaling pathways, such as the pathway for greening in response to light (see Figure 39.4). Cells use Ca^{2+} as a second messenger in pathways triggered by both G protein-coupled receptors and receptor tyrosine kinases.

Although cells always contain some Ca^{2+}, this ion can function as a second messenger because its concentration in the cytosol is normally much lower than the concentration outside the cell **(Figure 11.13)**. In fact, the level of Ca^{2+} in the blood and extracellular fluid of an animal is often more than 10,000 times higher than that in the cytosol. Calcium ions are actively transported out of the cell and are actively imported from the cytosol into the endoplasmic reticulum (and, under some conditions, into mitochondria and chloroplasts) by various protein pumps. As a result, the calcium concentration in the ER is usually much higher than that in the cytosol. Because the cytosolic calcium level is low, a small change in absolute numbers of ions represents a relatively large percentage change in local calcium concentration.

Figure 11.13

```
    The maintenance of calcium ion concentrations in an animal
cell.
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The maintenance of calcium ion concentrations in an animal cell. C a 2 plus pumps are in the plasma membrane, mitochondrion membrane, and E R. A T P drives these pumps.

The Ca^{2+} concentration in the cytosol is usually much lower (light green) than in the extracellular fluid and ER (dark green). Protein pumps in the plasma membrane and the ER membrane, driven by ATP, move Ca^{2+} from the cytosol into the extracellular fluid and into the lumen of the ER. Mitochondrial pumps, driven by chemiosmosis (see Concept 9.4), move Ca^{2+} into mitochondria when the calcium level in the cytosol rises significantly.

In response to a signal relayed by a signal transduction pathway, the cytosolic calcium level may rise, usually by a mechanism that releases Ca^{2+} from the cell's ER. The pathways leading to calcium release involve two other second messengers, **inositol trisphosphate** (IP_3) and **diacylglycerol (DAG)**. These two messengers are produced by cleavage of a certain kind of phospholipid in the plasma membrane. **Figure 11.14** shows the complete picture of how a signal causes IP_3 to stimulate the release of calcium from the ER.

Because IP3 acts before calcium in these pathways, calcium could be considered a "*third* messenger." However, scientists use the term *second messenger* for all small, nonprotein components of signal transduction pathways.

Figure 11.14

Calcium and IP3 in signaling pathways.

Calcium and I P 3 in signaling pathways.

- A signaling molecule binds to a receptor, leading to activation of phospholipase C.
- Phospholipase C cleaves a plasma membrane phospholipid called P I P 2 into DAG and I P 3.
- DAG functions as a second messenger in other pathways. DAG stays in the membrane while I P 3 moves into the cytosol.
- I P 3 quickly diffuses through the cytosol and binds to an I P 3-gated calcium channel in the E R membrane, causing it to open.
- Calcium ions flow out of the E R (down their concentration gradient), raising the C a 2 plus level in the cytosol. C a 2 plus acts as a second messenger.
- The calcium ions activate the next protein in one or more signaling pathways. The various proteins activated trigger cellular responses.

Calcium ions (Ca2+) and inositol trisphosphate (IP3) function as second messengers in many signal transduction pathways. In this figure, the process is initiated by the binding of a signaling molecule to a G protein-coupled receptor. A receptor tyrosine kinase could also initiate this pathway by activating phospholipase C.

MAKE CONNECTIONS *Explain the difference in function (in terms of ion transport) between the*Ca2+*pump in**Figure 11.13**and the*Ca2+*channel protein shown here. (See*Figure 7.17*.)**

*For suggested answer, see*Appendix A*.**

Concept 11.4: Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities

We now take a closer look at the cell's subsequent response to an extracellular signal—what some researchers call the "output response." What is the nature of the final step in a signaling pathway?

Nuclear and Cytoplasmic Responses

Ultimately, a signal transduction pathway leads to the regulation of one or more cellular activities. The response may occur in the nucleus or in the cytoplasm of the cell.

Many signaling pathways ultimately regulate protein synthesis, usually by turning specific genes on or off in the nucleus. Like an activated steroid receptor (see Figure 11.9), the final activated molecule in a signaling pathway may function as a transcription factor. **Figure 11.15** shows an example in which a signaling pathway activates a transcription factor that turns a gene on: The response to this growth factor signal is transcription, the synthesis of one or more specific mRNAs, which will be translated in the cytoplasm into specific proteins. In other cases, the transcription factor might regulate a gene by turning it off. Often, a transcription factor regulates several different genes.

Figure 11.15

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    Nuclear responses to a signal: the activation of a specific
gene by a growth factor.
```

Reception, transduction and response in the form of m R N A transcription illustrated in reaction to the binding of a growth factor to a receptor in the plasma membrane.

Reception: a growth factor in the extracellular fluid binds to a receptor in the plasma membrane, leading to transduction.

Transduction: a phosphorylation cascade occurs in the cytoplasm, leading to:

Response: An inactive transcription factor in the cell nucleus is activated (phosphorylated), causing transcription of a gene (segment of DNA) into mRNA.

This diagram shows a typical signaling pathway that leads to regulation of gene activity in the cell nucleus. The initial signaling molecule, in this case a growth factor, triggers a phosphorylation cascade, as in Figure 11.10. (The ATP molecules and phosphate groups are not shown.) Once phosphorylated, the last kinase in the sequence enters the nucleus and activates a transcription factor, which stimulates transcription of a specific gene (or genes). The resulting mRNAs then direct the synthesis of a particular protein.

Animation

Nuclear Response: Activating a Gene

image

Play

image

Play

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Sometimes a signaling pathway may regulate the *activity* of proteins rather than causing their synthesis by activating gene expression. This directly affects proteins that function outside the nucleus. For example, a signal may cause the opening or closing of an ion channel in the plasma membrane or a change in the activity of a metabolic enzyme. As we have seen, the response of liver cells to the hormone epinephrine helps regulate cellular energy metabolism by affecting the activity of an enzyme. The final step in the epinephrine signaling pathway activates the enzyme that catalyzes the breakdown of glycogen. **Figure 11.16** shows the complete pathway leading to the release of glucose 1-phosphate molecules from glycogen. Notice that as each molecule is activated, the response is amplified, a subject we'll return to shortly.

Figure 11.16

```
    Cytoplasmic response to a signal: the stimulation of
glycogen breakdown by epinephrine (adrenaline).
```

Cytoplasmic response to a signal.

Reception: epinephrine binds to G protein coupled receptor (1 molecule).

Transduction: Inactive G protein is activated to active G protein (10 to the power of 2 molecules. Inactive adenylyl cyclase is activated to active adenylyl cyclase (10 to the power of 2 molecules). ATP goes to Cyclic AMP (10 to the power of 4 molecules). Inactive protein kinase A is activated to active protein kinase A (10 to the power of 4 molecules). Inactive phosphorylase kinase is activated to active phosphorylase kinase (10 to the power of 5 molecules). Inactive glycogen phosphorylase is activated to active glycogen phosphorylase (10 to the power of 6 molecules).

Response: Glycogen goes to Glucose 1-phosphate (10 to the power of 8 molecules).

In this signaling system, the hormone epinephrine acts through a G protein-coupled receptor to activate a succession of relay molecules, including cAMP and two protein kinases (see also Figure 11.12). The final protein activated is the enzyme glycogen phosphorylase, which uses inorganic phosphate to release glucose

monomers from glycogen in the form of glucose 1-phosphate molecules. This pathway amplifies the hormonal signal: One receptor protein can activate approximately 100 molecules of G protein, and each enzyme in the pathway, once activated, can act on many molecules of its substrate, the next molecule in the cascade. The number of activated molecules given for each step is approximate.

VISUAL SKILLS *In the figure, how many glucose*1-phosphate*molecules are released in response to one signaling molecule? Calculate the factor by which the response is amplified in going from each step to the next.*

*For suggested answer, see*Appendix A.**

Animation

Cytoplasmic Response: Glycogen Breakdown

image

Play

image

Play

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Signal receptors, relay molecules, and second messengers participate in a variety of pathways, leading to both nuclear and cytoplasmic responses, including cell division. Malfunctioning of growth factor pathways like the one in Figure 11.15 can contribute to abnormal cell division and the development of cancer, as we'll see in Concept 18.5.

Regulation of the Response

Whether the response occurs in the nucleus or in the cytoplasm, it is not simply turned "on" or "off." Rather, the extent and specificity of the response are regulated in multiple ways. Here we'll consider four aspects of this regulation. First, as mentioned earlier, signaling pathways generally amplify the cell's response to a single signaling event. The degree of amplification depends on the function of the specific molecules in the pathway. Second, the many steps in a multistep pathway provide control points at which the cell's response can be further regulated, contributing to the specificity of the response and allowing coordination with other signaling pathways. Third, the overall efficiency of the response is enhanced by the presence of proteins known as scaffolding proteins. Finally, a crucial point in regulating the response is the termination of the signal.

Signal Amplification

Elaborate enzyme cascades amplify the cell's response to a signal. At each catalytic step in the cascade, the number of activated products can be much greater than in the preceding step. For example, in the epinephrine-triggered pathway in Figure 11.16, each adenylyl cyclase molecule catalyzes the formation of 100 or so cAMP molecules, each molecule of protein kinase A phosphorylates about 10 molecules of the next kinase in the pathway, and so on. The amplification effect stems from the fact that these proteins persist in their active form long enough to process multiple molecules of substrate before they become inactive again. As a result of the signal's amplification, a small number of epinephrine molecules binding to receptors on the surface of a liver cell or muscle cell can lead to the release of hundreds of millions of glucose molecules from glycogen.

The Specificity of Cell Signaling and Coordination of the Response

Consider two different cells in your body—a liver cell and a heart muscle cell, for example. Both are in contact with your bloodstream and are therefore constantly exposed to many different hormone molecules, as well as to local regulators secreted by nearby cells. Yet the liver cell responds to some signals but ignores others, and the same is true for the heart cell. And some kinds of signals trigger responses in both cells—but different responses. For instance, epinephrine stimulates the liver cell to break down glycogen, but the main response of the heart cell to epinephrine is contraction, leading to a more rapid heartbeat. How do we account for this difference?

The explanation for the specificity exhibited in cellular responses to signals is the same as the basic explanation for virtually all differences between cells: Because different kinds of cells turn on different sets of genes, *different kinds of cells have different collections of proteins*. The response of a cell to a signal depends on its particular collection of signal receptor proteins, relay proteins, and proteins needed to carry out the response. A liver cell, for example, is poised to respond appropriately to epinephrine by having the proteins listed in Figure 11.16 as well as those needed to manufacture glycogen.

Thus, two cells that respond differently to the same signal differ in one or more proteins that respond to the signal. Also, within some cells, there are more complex pathways that branch or converge, as shown in **Figure 11.17**. Notice, though, that these different pathways still have some molecules in common. For example, cells A, B, and C all use the same receptor protein for the red signaling molecule; differences in other proteins account for their differing responses. In cell D, a different receptor protein is used for the same signaling molecule, leading to yet another response. In cell B, a pathway triggered by one signal diverges to produce two responses; such branched pathways often involve receptor tyrosine kinases (which can activate multiple relay proteins) or second messengers (which can regulate numerous proteins). In cell

C, two pathways triggered by separate signals converge to modulate a single response. Branching of pathways and “cross-talk” (interaction) between pathways are important in regulating and coordinating a cell’s responses to information coming in from different sources in the body. (You’ll learn more about this coordination in Concept 11.5.) Moreover, the use of some of the same proteins in more than one pathway allows the cell to economize on the number of different proteins it must make.

Figure 11.17

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The specificity of cell signaling.
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Specificity of cell signaling shown in Cells A, B, C and D.

The particular proteins a cell possesses determine what signaling molecules it responds to and the nature of the response. The four cells in these diagrams respond to the same signaling molecule (red) in different ways because each has a different set of proteins (purple and teal). Note, however, that the same kinds of molecules can participate in more than one pathway.

Cell A: Pathway leads to a single response. Signaling molecule in the extracellular fluid binds to receptor in the plasma membrane, triggering a pathway of relay molecules inside the cell that lead to a single Response 1.

. Cell B: Pathway branches to two responses. The same signal molecule shown in cell A binds to the same receptor in a cell membrane, but this time the relay molecule pathway branches, leading to two different responses (responses 2 and 3).

Cell C: Cross-talk occurs between two pathways. Signal molecule from cells A and B binds again to the same membrane receptor, triggering response 4 following a relay molecule pathway inside the cell. This time, there is also another receptor in the plasma membrane that binds a different signal molecule; the result of this pathway in activation of inhibition of the pathway leading to response 4 in the cell.

Cell D: Different receptor leads to a different response. The same signal molecule shown in cells A, B and C binds to a different receptor in the plasma membrane, activating a relay molecule pathway that leads to response 5.

VISUAL SKILLS *Study the signaling pathway shown in*Figure 11.14*, and explain how the situation pictured for cell B in***Figure 11.17**could apply to that pathway.*

*For suggested answer, see*Appendix A.*

Instructors: The Scientific Skills Exercise “Using Experiments to Test a Model” can be assigned in **Mastering Biology**. It explores the cellular response of a yeast cell to the signal initiated by a mating factor

from a cell of the opposite mating type.

Signaling Efficiency: Scaffolding Proteins and Signaling Complexes

The illustrations of signaling pathways in Figure 11.17 (as well as diagrams of other pathways in this chapter) are greatly simplified. The diagrams show only a few relay molecules and, for clarity's sake, display these molecules spread out in the cytosol. If this were true in the cell, signaling pathways would operate very inefficiently because most relay molecules are proteins, and proteins are too large to diffuse quickly through the viscous cytosol. How does a given protein kinase, for instance, find its protein substrate?

In many cases, the efficiency of signal transduction is apparently increased by the presence of **scaffolding proteins**, large relay proteins to which several other relay proteins are simultaneously attached **(Figure 11.18)**. Researchers have found scaffolding proteins in brain cells that *permanently* hold together networks of signaling pathway proteins at synapses. This hardwiring enhances the speed and accuracy of signal transfer between cells because the rate of protein-protein interaction is not limited by diffusion. Furthermore, in some cases the scaffolding proteins themselves may directly activate relay proteins.

Figure 11.18

```
A scaffolding protein.
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In cell A, a pathway leads to a single response.

The scaffolding protein shown here (orange) simultaneously binds to a specific activated membrane receptor and three different protein kinases. This physical arrangement facilitates signal transduction by these molecules.

The cell has a receptor in the membrane. A signaling molecule binds with the receptor outside the membrane. Relay molecules inside the membrane lead to Response 1.

The importance of the relay proteins that serve as points of branching or intersection in signaling pathways is highlighted by the problems arising when these proteins are defective or missing. For instance, in an inherited disorder called Wiskott-Aldrich syndrome (WAS), the absence of a single relay protein leads to such diverse effects as abnormal bleeding, eczema, and a predisposition to infections and leukemia. These symptoms are thought to arise primarily from the absence of the protein in cells of the immune system. By studying normal cells, scientists found that the WAS protein is located just beneath the immune cell surface. The protein interacts both with microfilaments of the cytoskeleton and with several different components of signaling pathways that relay information from the cell surface, including pathways regulating immune cell proliferation. This multifunctional relay protein is thus both a branch point and an important intersection

point in a complex signal transduction network that controls immune cell behavior. When the WAS protein is absent, the cytoskeleton is not properly organized and signaling pathways are disrupted, leading to the WAS symptoms.

Termination of the Signal

In the interest of keeping Figure 11.17 simple, we did not show the *inactivation* mechanisms that are an essential aspect of any cell-signaling pathway. For a cell of a multicellular organism to remain capable of responding to incoming signals, each molecular change in its signaling pathways must last only a short time. As we saw in the cholera example, if a signaling pathway component becomes locked into one state, whether active or inactive, consequences for the organism can be serious.

The ability of a cell to receive new signals depends on reversibility of the changes produced by prior signals. The binding of signaling molecules to receptors is reversible. As the external concentration of signaling molecules falls, fewer receptors are bound at any given moment, and the unbound receptors revert to their inactive form. The cellular response occurs only when the concentration of receptors with bound signaling molecules is above a certain threshold. When the number of active receptors falls below that threshold, the cellular response ceases. Then, by a variety of means, the relay molecules return to their inactive forms: The GTPase activity intrinsic to a G protein hydrolyzes its bound GTP; the enzyme phosphodiesterase converts cAMP to AMP; protein phosphatases inactivate phosphorylated kinases and other proteins; and so forth. As a result, the cell is soon ready to respond to a fresh signal.

BioFlix® Animation

Mechanism of Hormone Action: Second Messenger cAMP

image

Play

image

Play

In this section, we explored the complexity of signaling initiation and termination in a single pathway, and we saw the potential for pathways to intersect with each other. In the next section, we'll consider one especially important network of interacting pathways in the cell.

Concept 11.5: Apoptosis requires integration of multiple cell-signaling pathways

When signaling pathways were first discovered, they were thought to be linear, independent pathways. Our understanding of cellular communication has benefited from the realization that signaling pathway components interact with each other in various ways. For a cell to carry out the appropriate response, cellular proteins often must integrate multiple signals. Let's consider an important cellular process—cellular suicide—as an example.

Cells that are infected, are damaged, or have reached the end of their functional life span often undergo "programmed cell death" **(Figure 11.19)**. The best-understood type of this controlled cell suicide is **apoptosis** (from the Greek, meaning "falling off," and used in a classic Greek poem to refer to leaves falling from a tree). During this process, cellular agents chop up the DNA and fragment the organelles and other cytoplasmic components. The cell shrinks and becomes lobed (a change called "blebbing"), and the cell's parts are packaged up in vesicles that are engulfed and digested by specialized scavenger cells, leaving no trace. Apoptosis protects neighboring cells from damage that they would otherwise suffer if a dying cell merely leaked out all its contents, including its many digestive enzymes.

Figure 11.19

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Apoptosis of a human white blood cell.
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The cell on the left is more spherical while the cell on the right is covered in lobes. The scale is 2 micrometers.

On the left is a normal white blood cell, while on the right is a white blood cell undergoing apoptosis. The apoptotic cell is shrinking and forming lobes ("blebs"), which eventually are shed as membrane-bounded cell fragments (colorized SEMs).

The signal that triggers all of the complex events that occur during apoptosis can come from either outside or inside the cell. Outside the cell, signaling molecules released from other cells can initiate a signal transduction pathway that activates the genes and proteins responsible for carrying out cell death. Within a cell whose DNA has been irretrievably damaged, a series of protein-protein interactions can pass along a signal that similarly triggers cell death. Considering some examples of apoptosis can help us to see how signaling pathways are integrated in cells.

Apoptosis in the Soil Worm *Caenorhabditis elegans*

The molecular mechanisms of apoptosis were worked out by researchers studying embryonic development of a small soil worm, a nematode called *Caenorhabditis elegans*. Because the adult worm has only about 1,000 cells, the researchers were able to work out the entire ancestry of each cell. The timely suicide of cells occurs exactly 131 times during normal development of *C. elegans*, at precisely the same points in the cell lineage of each worm. In worms and other species, apoptosis is triggered by signals that activate a cascade of "suicide" proteins in the cells destined to die.

Genetic research on *C. elegans* initially revealed two key apoptosis genes, called Ced-3 and Ced-4 (*ced* stands for "cell death"), which encode proteins essential for apoptosis. The proteins are called Ced-3 and Ced-4, respectively. These and most other proteins involved in apoptosis are continually present in cells, but in inactive form; thus, regulation in this case occurs at the level of protein activity rather than through gene activity and protein synthesis. In *C. elegans*, a protein in the outer mitochondrial membrane, called Ced-9 (the product of the ced-9 gene), serves as a master regulator of apoptosis, acting as a brake in the absence of a signal promoting apoptosis **(Figure 11.20a)**. When a death signal is received by the cell, signal transduction involves a change in Ced-9 that disables the brake, and the apoptotic pathway activates proteases and nucleases, enzymes that cut up the proteins and DNA of the cell. The main proteases of apoptosis are called *caspases*; in the nematode, the chief caspase is the Ced-3 protein.

Figure 11.20

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Molecular basis of apoptosis in *C. elegans*.
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Molecular basis of apoptosis in C elegans.

Part a. No death signal. As long as C e d-9, located in the outer mitochondrial membrane, is active, apoptosis is inhibited, and the cell remains alive. Receptor in plasma membrane for death-signaling molecule is inactivated (open). C e d-9 protein in outer mitochondrial membrane is active, inhibiting C e d-4 activity. C e d-3 also inactive.

Part b. Death signal. When a cell receives a death signal, a series of proteins is activated. This triggers a cascade of reactions eventually leading to cell death. When a cell receives a death signal, C e d-9 is inactivated, relieving its inhibition of C e d-4. Active C e d-4 activates C e d-3, a protease, which triggers a cascade of reactions leading to the activation of nucleases and other proteases. The action of these enzymes causes the changes seen in apoptotic cells and eventually leads to cell death. Death-signaling molecule binds to plasma membrane receptor, triggering C e d-9 (in the outer mitochondrial membrane) to become inactive.

This leads to an activation cascade where C e d-4 and C e d-3 become active, triggering other proteases and nucleases. The cell forms blebs (lobes) in the plasma membrane.

Three proteins, Ced-3, Ced-4, and Ced-9, are critical to apoptosis and its regulation in the nematode. Apoptosis is more complicated in mammals but involves proteins similar to those in *C. elegans*.

Apoptotic Pathways and the Signals That Trigger Them

In humans and other mammals, several different pathways, involving about 15 different caspases, can carry out apoptosis. The pathway that is used depends on the type of cell and on the particular signal that initiates apoptosis. One major pathway involves certain mitochondrial proteins that are triggered to form molecular pores in the mitochondrial outer membrane, causing it to leak and release other proteins that promote apoptosis. Perhaps surprisingly, these latter include cytochrome c, which functions in mitochondrial electron transport in healthy cells (see Figure 9.14) but acts as a cell death factor when released from mitochondria. The process of mitochondrial apoptosis in mammals uses proteins similar to the nematode proteins Ced-3, Ced-4, and Ced-9. These can be thought of as relay proteins capable of transducing the apoptotic signal.

At key gateways into the apoptotic program, relay proteins integrate signals from several different sources and can send a cell down an apoptotic pathway. Often, the signal originates outside the cell, like the death-signaling molecule depicted in **Figure 11.20b**, which presumably was released by a neighboring cell. When a death-signaling ligand occupies a cell-surface receptor, this binding leads to activation of caspases and other enzymes that carry out apoptosis, without involving the mitochondrial pathway. This process of signal reception, transduction, and response is similar to what we have discussed throughout this chapter. In a twist on the classic scenario, two other types of alarm signals that can lead to apoptosis originate from *inside* the cell rather than from a cell-surface receptor. One signal comes from the nucleus, generated when the DNA has suffered irreparable damage, and a second comes from the endoplasmic reticulum when excessive protein misfolding occurs. Mammalian cells make life-or-death “decisions” by somehow integrating the death signals and life signals they receive from these external and internal sources.

A built-in cell suicide mechanism is essential to development and maintenance in all animals. The similarities between apoptosis genes in nematodes and those in mammals, as well as the observation that apoptosis occurs in multicellular fungi and even in single-celled yeasts, indicate that the basic mechanism evolved early in the evolution of eukaryotes. In vertebrates, apoptosis is essential for normal development of the nervous system, for normal operation of the immune system, and for normal morphogenesis of hands and feet in humans and paws in other mammals **(Figure 11.21)**. The level of apoptosis between the developing digits is lower in the webbed feet of ducks and other water birds than in the nonwebbed feet of

land birds, such as chickens. In the case of humans, the failure of appropriate apoptosis can result in webbed fingers and toes.

Figure 11.21

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Effect of apoptosis during paw development in the mouse.
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Effect of apoptosis during paw development in the mouse, shown in 3 images.

In mice, humans, other mammals, and land birds, the embryonic region that develops into feet or hands initially has a solid, platelike structure. Apoptosis eliminates the cells in the interdigital regions, thus forming the digits. The embryonic mouse paws shown in these fluorescence light micrographs are stained so that cells undergoing apoptosis appear a bright yellowish green. Apoptosis of cells begins at the margin of each interdigital region (left), peaks as the tissue in these regions is reduced (middle), and is no longer visible when the interdigital tissue has been eliminated (right).

First image: no space between digits in mouse paw; interdigital tissue is present. Yellow-green layer along outermost section of the paw shows cells undergoing apoptosis.

Second image: some of the interdigital tissue has been lost; cells remaining between the digits are yellow-green, indicating that they are undergoing apoptosis.

Third image: Apoptosis is complete; clear spaces now evident between digits. Scale: 1 millimeters.

Significant evidence points to the involvement of apoptosis in certain degenerative diseases of the nervous system, such as Parkinson's disease and Alzheimer's disease. In Alzheimer's disease, an accumulation of aggregated proteins in neuronal cells activates an enzyme that triggers apoptosis, resulting in the loss of brain function seen in these patients. Furthermore, cancer can result from a failure of cell suicide; some cases of human melanoma, for example, have been linked to faulty forms of the human version of the *C. elegans* Ced-4 protein. It is not surprising, therefore, that the signaling pathways feeding into apoptosis are quite elaborate. After all, the life-or-death question is the most fundamental one imaginable for a cell.

This chapter has introduced you to many of the general mechanisms of cell communication, such as ligand binding, protein-protein interactions and shape changes, cascades of interactions, and protein phosphorylation. Throughout your study of biology, you will encounter numerous examples of cell signaling.